



Engineering Project Management & Finance (UESTC 3031 & UESTCHN 3012)

Design For Manufacturing (Part 4)

Dr. Amir Parnianifard

Email: Amir.Parnianifard@glasgow.ac.uk

Outline

- **Part 1:** Introduction to DFM & Design for Sustainability
- **Part 2:** Quality Control & Cost of Quality
- **Part 3:** Robust Manufacturing Design
- **Part 4:** 6-Sigma & Process Capability



Six-Sigma (6σ)

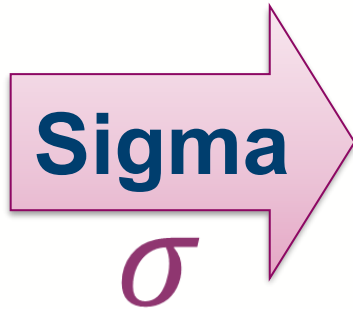


Six-Sigma: Contents:

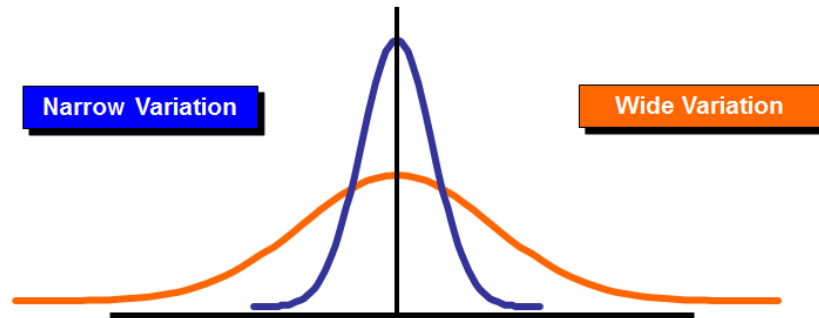
- Six Sigma : An Overview
- What is Six Sigma?
- Why Six Sigma?
- Six Sigma Phases : Define, Measure, Analyze, Improve and Control
- Tools and Key Roles for Six Sigma



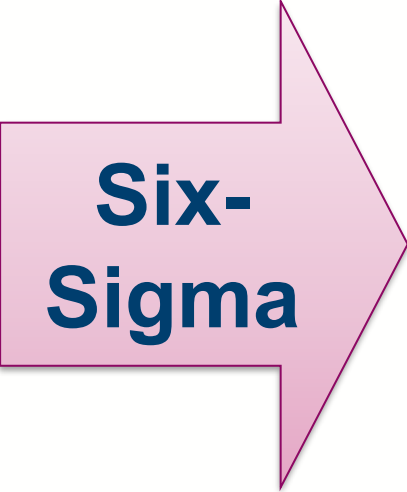
What is Sigma ?



A term used in statistics to represent standard deviation, an indicator of the degree of variation in a set of a process.



What is Six-Sigma ?



**Six-
Sigma**

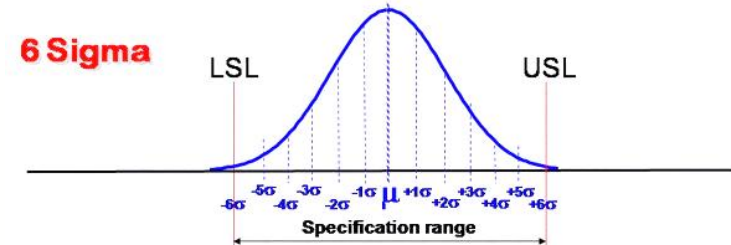
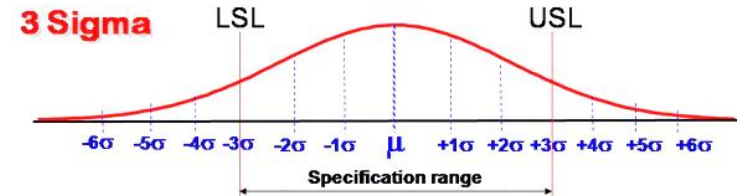
A statistical concept that measures a process in terms of defects – at the six-sigma level, there 3.4 defects per million opportunities.

Goal: as perfect as practically possible

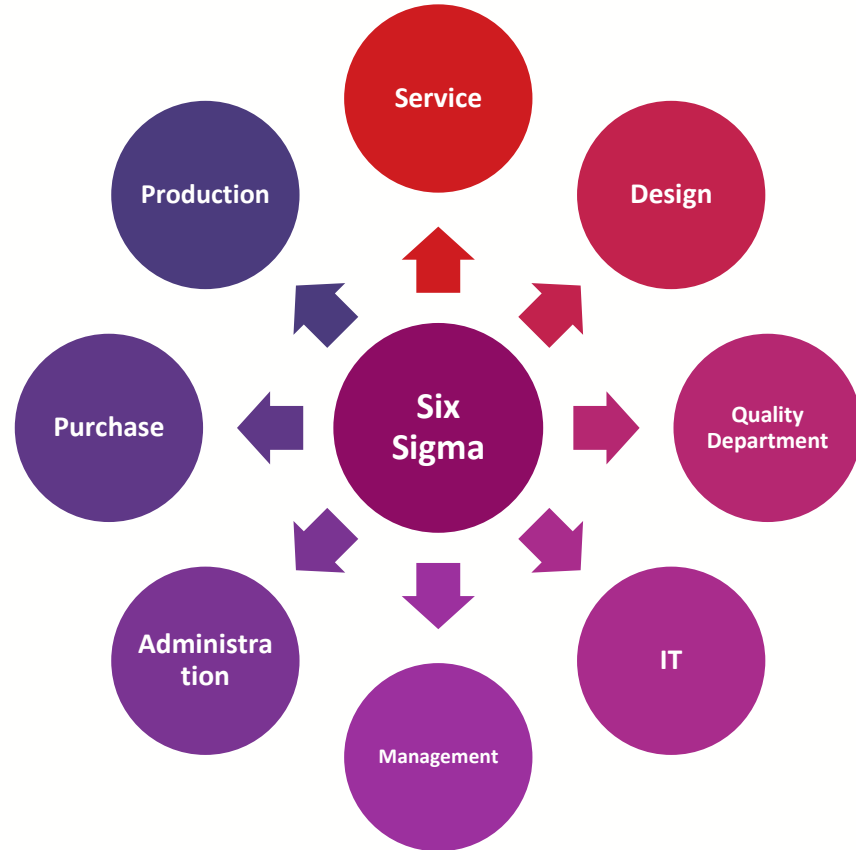
A methodology and a symbol of quality

Sigma Levels

Sigma Level (Process Capability)	Defects per Million Opportunities
2	308,537
3	66,807
4	6,210
5	233
6	3.4



Field of Application



Cost of Poor Quality

- Cost of scrap.
- Cost of rework.
- Cost of excessive cycle times and delays.
- Cost of business lost because customers are dissatisfied with your products or services.
- Cost of opportunities lost because you didn't have time or the resources to take advantage of them.

Critical-to-Quality (CTQ)

Elements of a process that significantly affect the output of that process. Identifying these elements is figuring out how to make improvements that can dramatically reduce costs and enhance quality.



Why Six Sigma?

Money

Quality

**Customer
Satisfaction**

**Competitive
Advantage**

Growth

Employee Pride

Why Six Sigma?

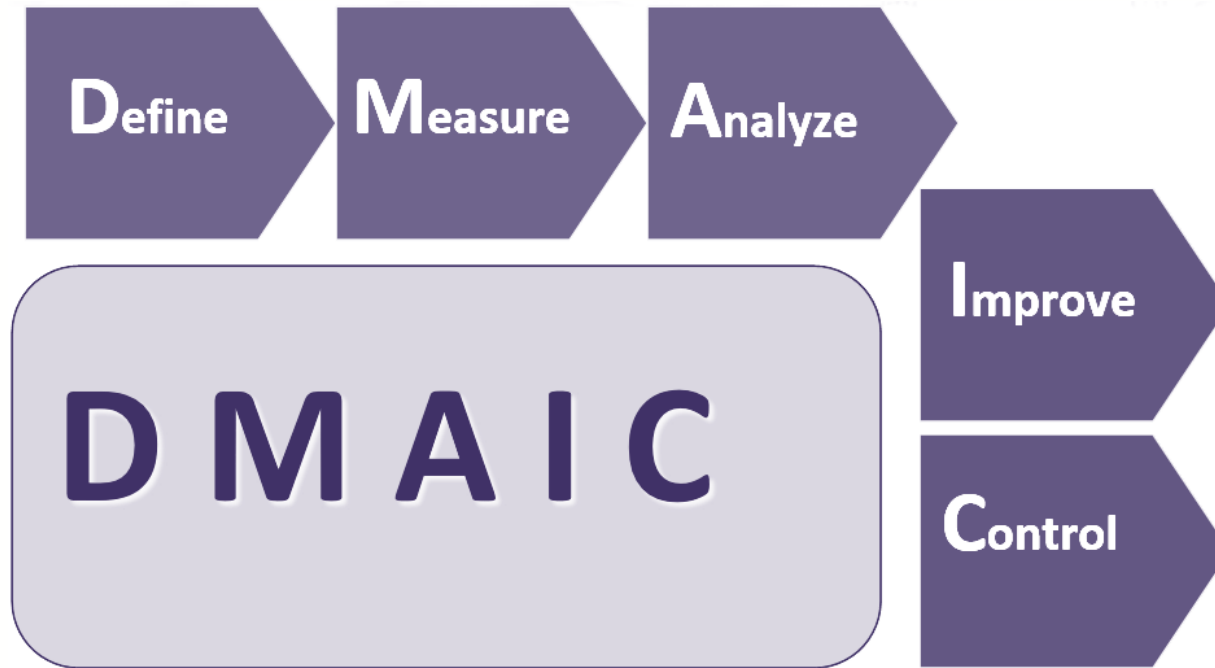
- At General Electric Company (GE), Six Sigma added more than \$2 billion to a company's net income in 1999 alone.
- Motorola saved more than \$15 billion in the first 10 years of its Six Sigma effort.
- AlliedSignal reports saving \$1,5 billion through Six Sigma.

Six Sigma is about practices that help you eliminate defects and always deliver products and services that meet customer specifications.

Some companies that implemented Six-Sigma:

- 3M
- Air Canada
- Amazon.Com
- Bank of America
- Boeing
- Caterpillar, Inc.
- Corning
- Dell
- DHL
- Eastman Kodak Company
- Ford
- General Electric
- GlaxoSmithKline
- HSBC Group
- LG Group
- Motorola
- Samsung Group
- Siemens AG
- McGraw-Hill Companies
- US Air Force, US Navy, US Marine Corps
- Vodafone
- Whirlpool
- Xerox

Six Sigma Phases



The DMAIC model is a **roadmap for Six Sigma**, used to **improve the quality of results** that company processes produce.

DMAIC

Define

Define the project goals and customer (internal and external) deliverables.

Measure

Measure the process to determine current performance.

Analyze

Analyze and determine the root cause(s) of the defects

DMAIC

Improve

Improve the process by eliminating defects.

Control

Monitor all improvements made to ensure sustainable changes and long-lasting results.

Tools for Six Sigma

Brainstorming Rules

1 | Don't Evaluate

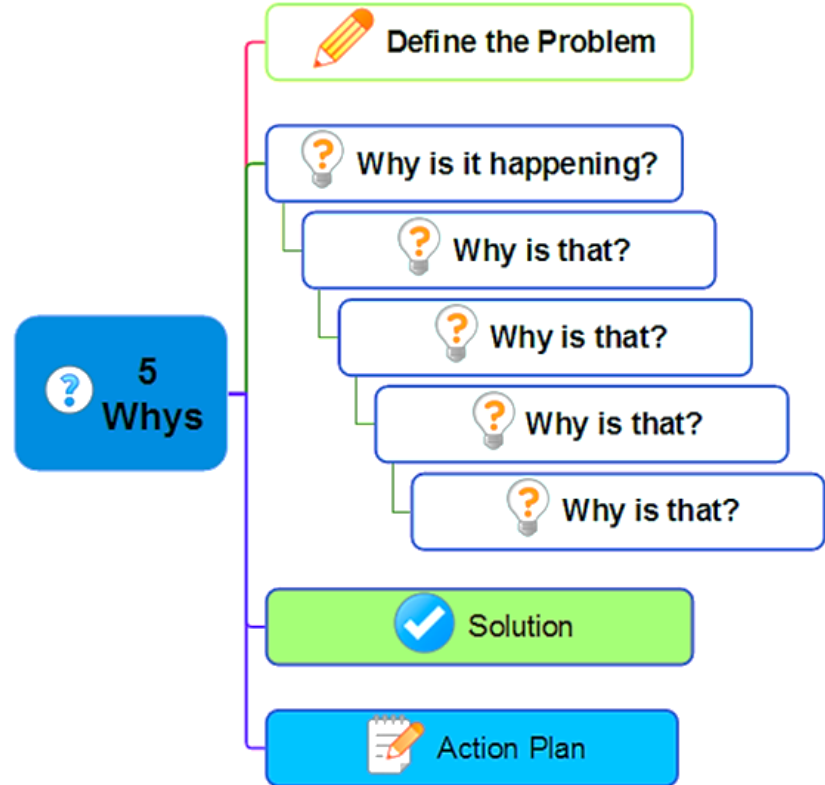


Quantity Needed | 2

3 | Keep Building

Wild Ideas Wanted | 4

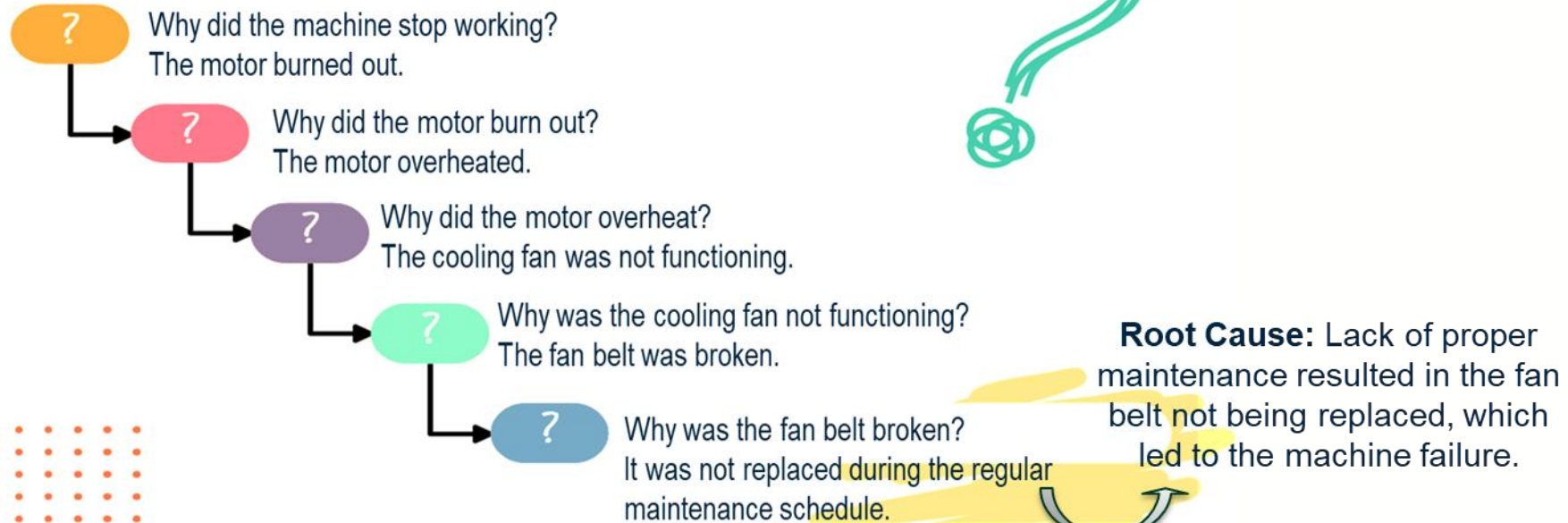
There are no other rules!



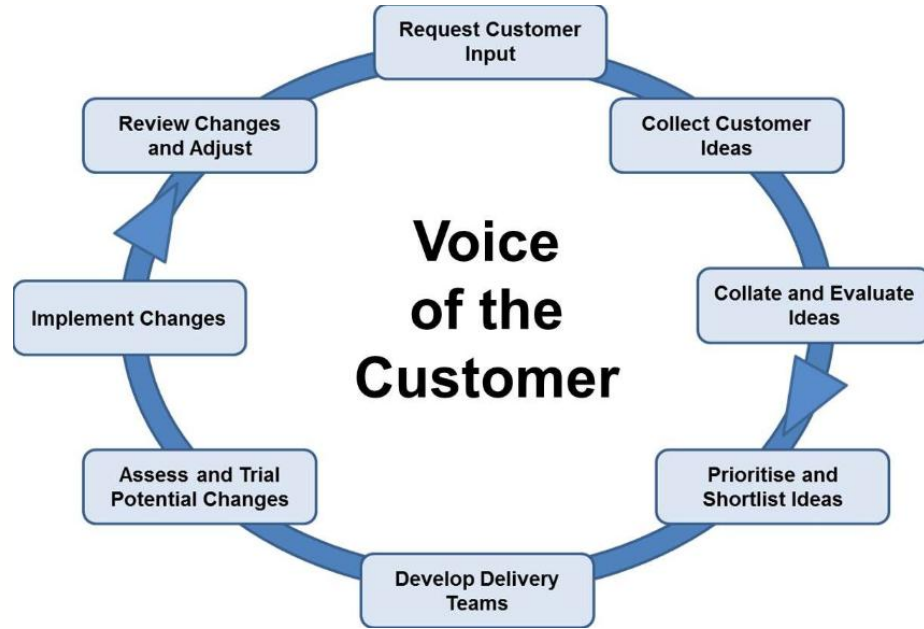
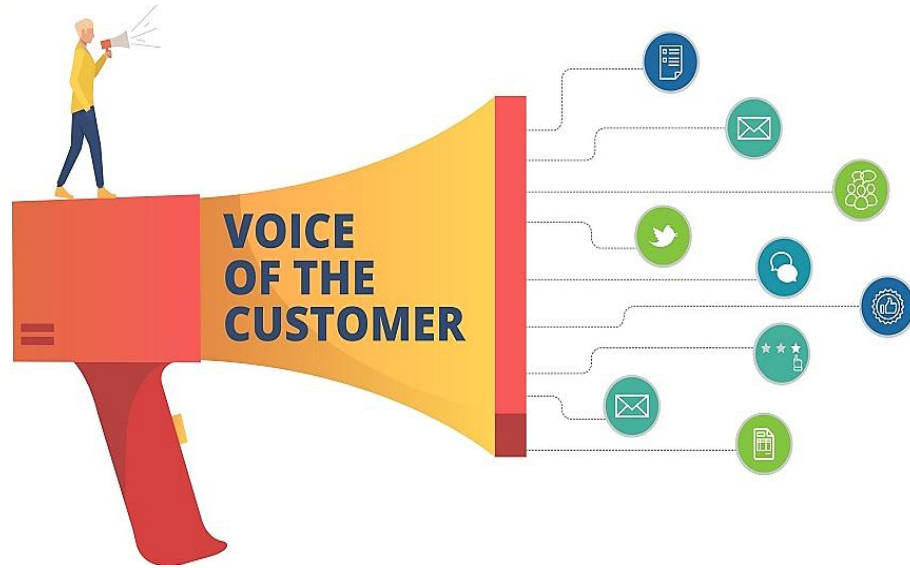
Tools for Six Sigma

Example: (5 Whys in problem solving)

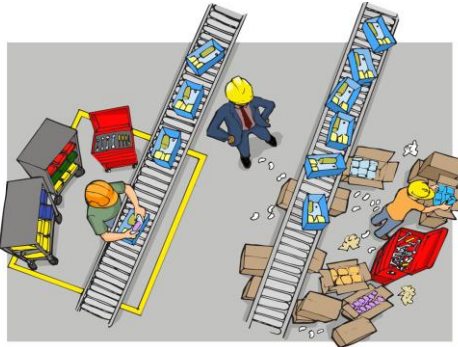
5 Why Method



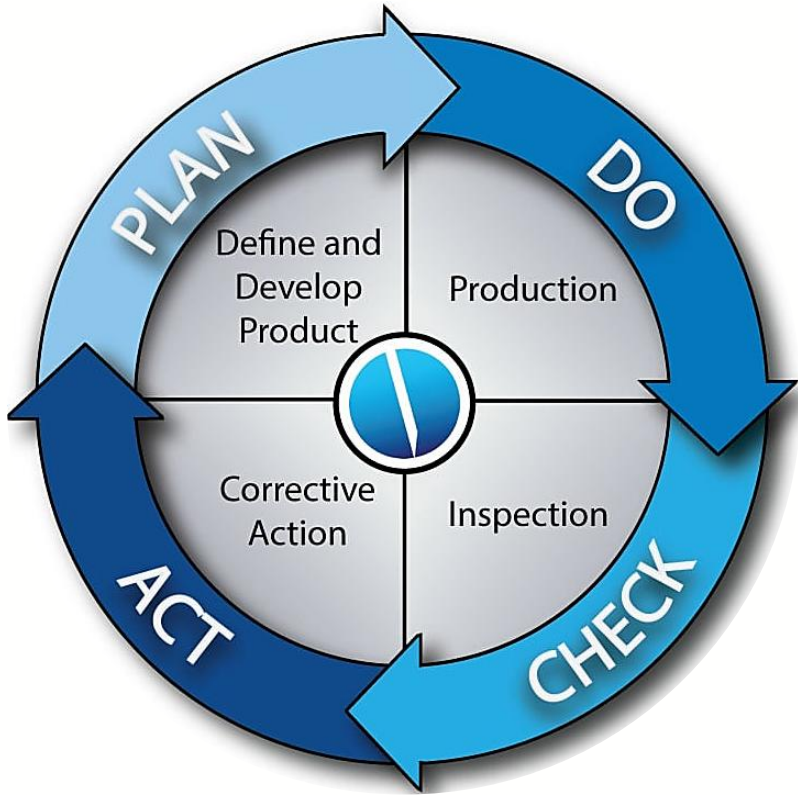
Tools for Six Sigma



Tools for Six Sigma



Tools for Six Sigma



Continuous Improvement with PDCA

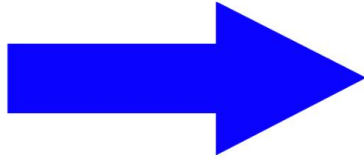


Tools for Six Sigma Mistake Proofing

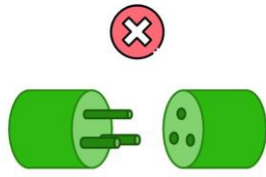
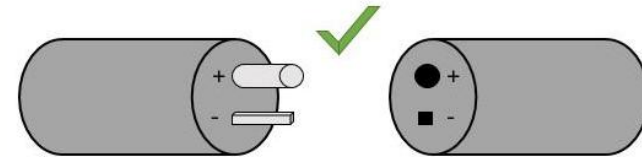
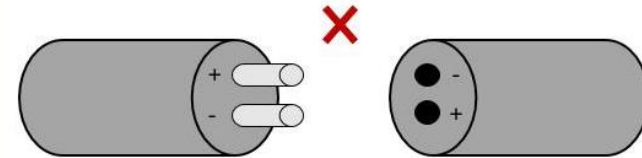
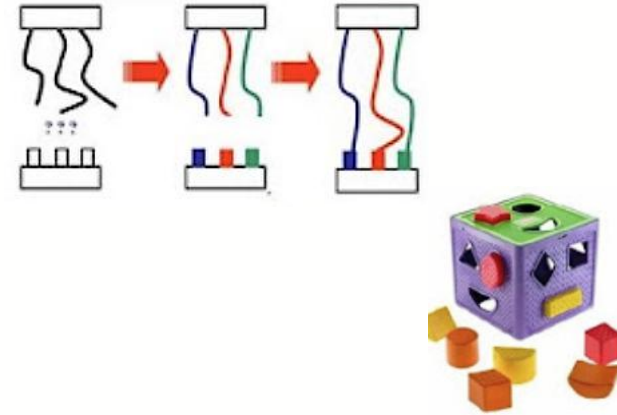
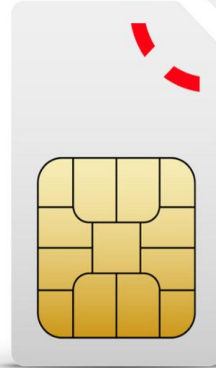
Confusion



Poka-yoke



Clarity



Key Roles for Six Sigma

There are five key roles for Six Sigma in manufacturing:



Executive Leadership

Champions

Master Black Belts

Black Belts

Green Belts

Process Capability



What is Process Capability?

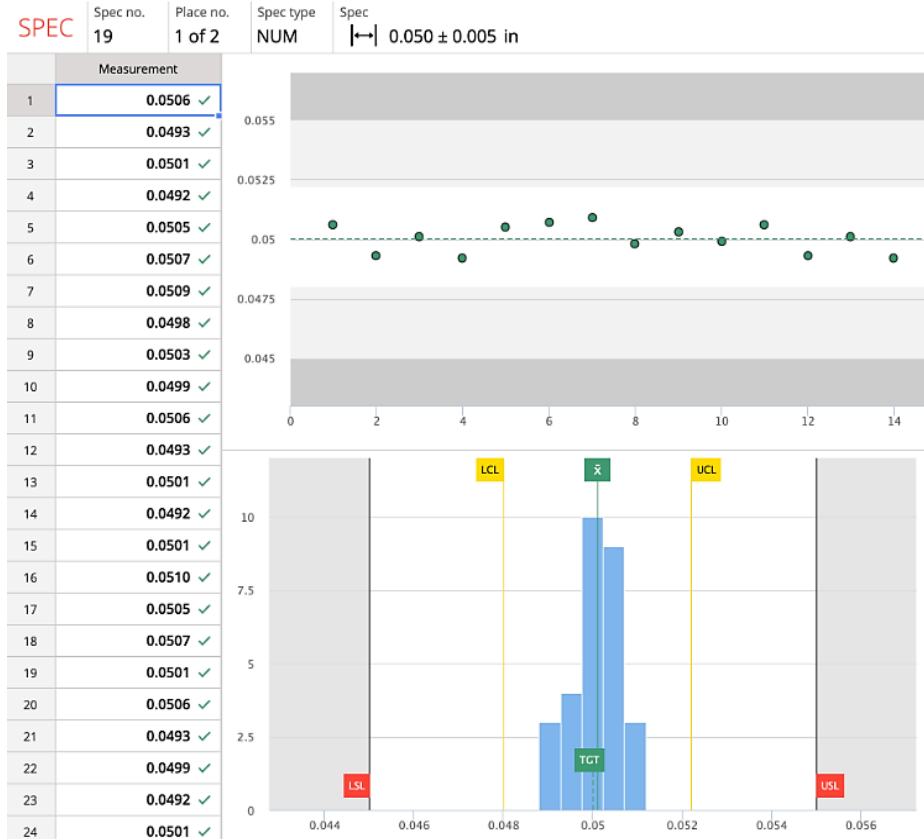
You will take some historical data, and extrapolate out to the future to answer the question:

Can I rely on this process to deliver good parts?



Definition:

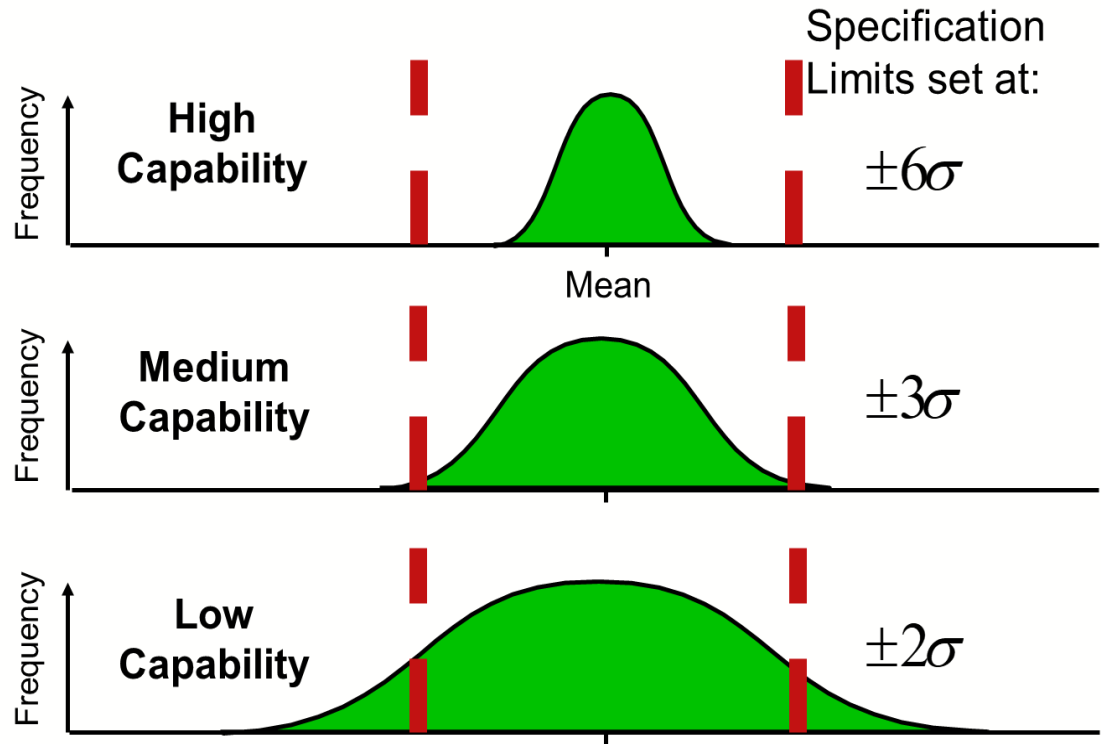
Process capability refers to the **measure/degree** to which a process can consistently **produce output** that **meets process specifications**.



Process Capability Overview

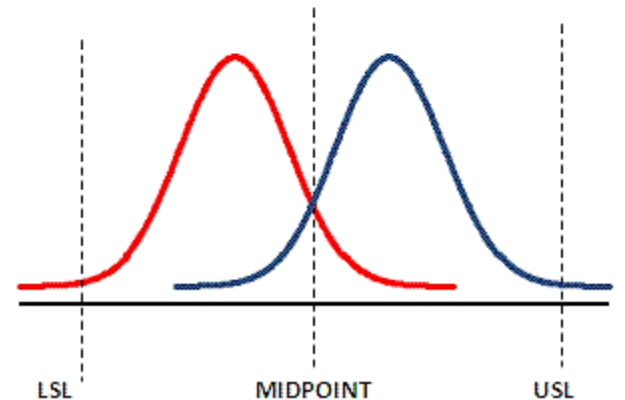
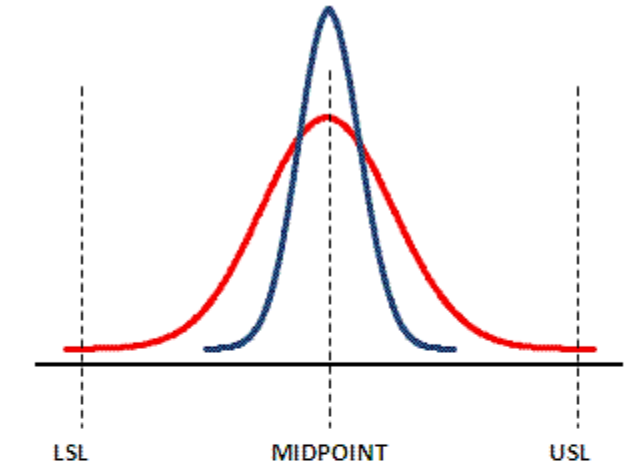


The goal of process improvement is to **shift the process center to the target value** and **reduce variation** so that results consistently meet customer specifications..



Process Capability Indices

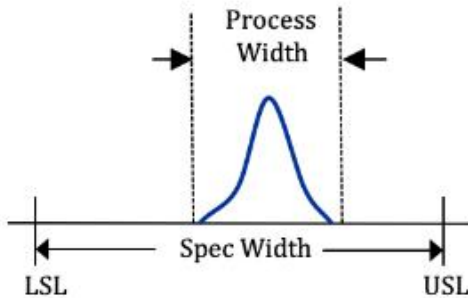
- ✓ There are **two main statistics** that can be used to measure the capability of a process (C_p and C_{pk}).
- ✓ Most capability indices estimates are valid only if the **sample size** used is "**large enough**". Large enough is generally thought to be about 50 independent data values.
- ✓ The C_p and C_{pk} statistics assume that the population of data values is **normally distributed**.
- ✓ μ and σ are the mean and standard deviation, respectively, of the normal data and USL and LSL are the upper and lower specification limits, respectively.



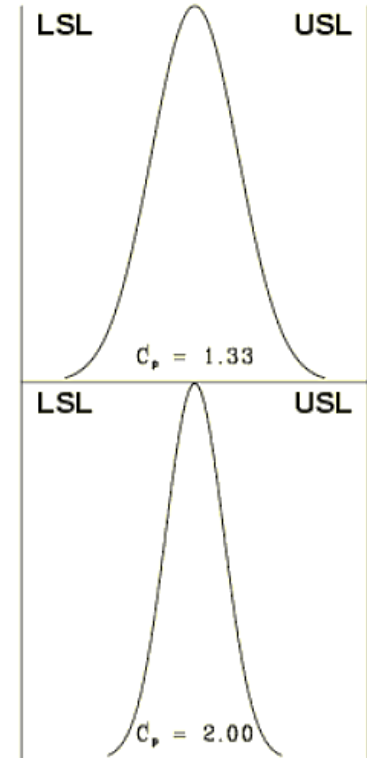
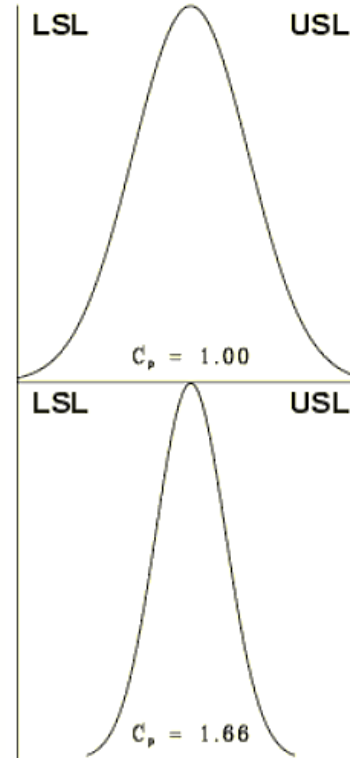
Process Capability Indices

$$C_p = \frac{USL - LSL}{6\sigma}$$

$$C_p = \frac{\text{Specification Width}}{\text{Process Width}}$$



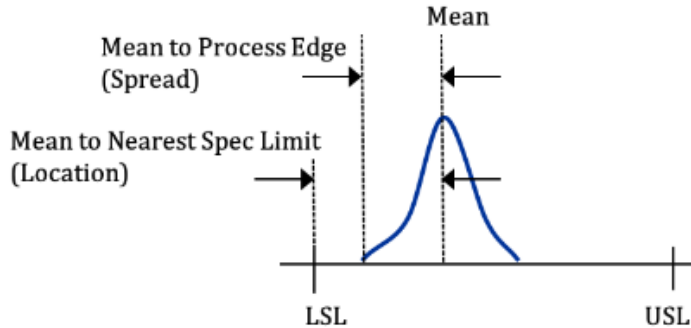
C_p accounts for only the spread (or variation) of the process.



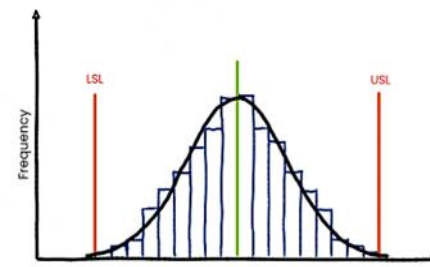
Process Capability Indices

$$C_{pk} = \min \left[\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right]$$

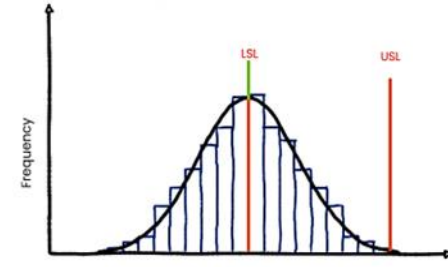
$$C_{pk} = \frac{\text{Distance from Mean to Nearest Spec Limit}}{\text{Distance from Mean to Process Edge}}$$



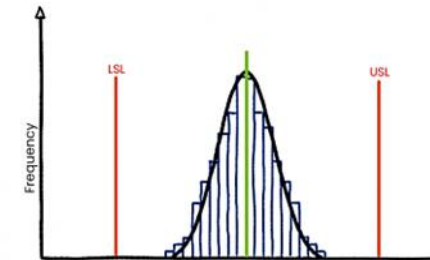
C_{pk} accounts for both the spread and location of the process.



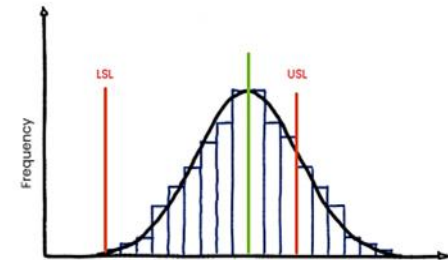
C_{pk} = 1



C_{pk} = 0



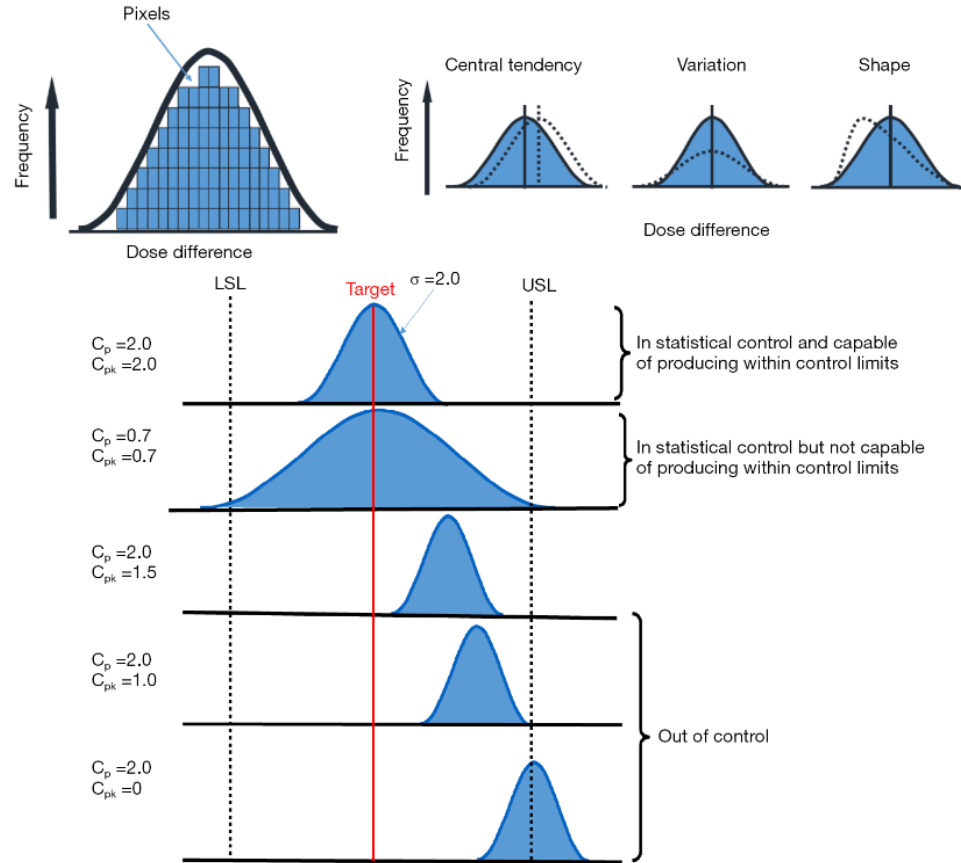
C_{pk} = 1.67



C_{pk} = 0.33

Process Capability Indices

Example:

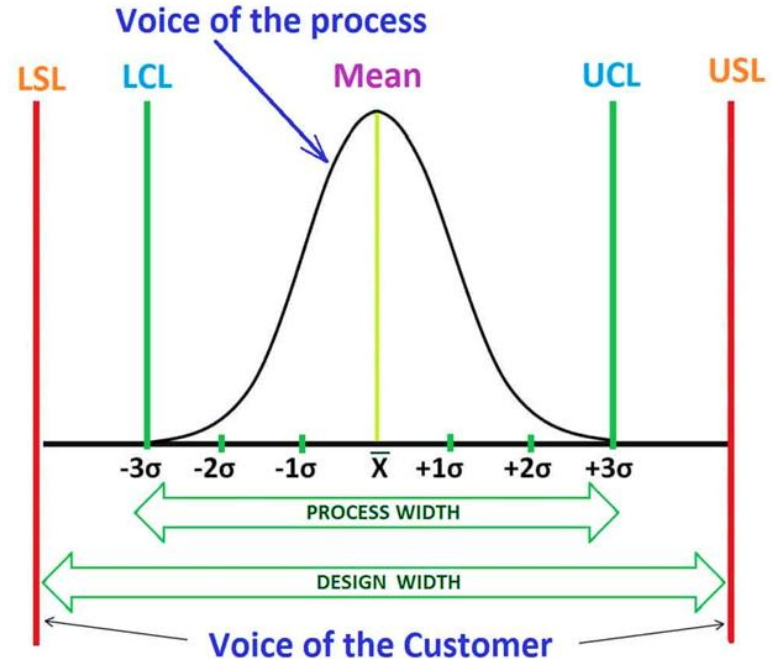


Process Capability Indices

Risk Assessment for Cpk Value:

- $C_{pk} < 1.00$: High risk of defects, the process is not capable.
- $1.00 \leq C_{pk} < 1.33$: Moderate risk of defects, improvement needed for better capability.
- $1.33 \leq C_{pk} < 1.67$: Low risk of defects, the process is capable but has some room for improvement.
- $C_{pk} \geq 1.67$: Minimal risk of defects, the process is highly capable.

Aim for a C_{pk} value of 2.00 or higher where possible. A C_{pk} of 2.00 implies that the process uses only 50% of the spec width, significantly **reducing the risk of defect**.



Process Capability Indices

Example:

A manufacturer produces a type of resistor that should have a resistance of 10 ohms, with a tolerance of ± 1 ohm. To ensure that the manufacturing process is capable of producing resistors within specifications, the manufacturer measures the resistance of a sample of resistors and calculates the Cp and Cpk indices. Assume that the sample of resistors has a mean resistance of 10.1 ohms and a standard deviation of 0.3 ohms. Discuss the obtained results.





Which of the following statements is true regarding process capability?

- A. Process capability is a measure of how efficient a process is in terms of time and resources.
- B. Process capability refers to the degree to which a process can consistently produce output that meets process specifications.
- C. Process capability is determined solely by the skill level of the workers involved in the process.
- D. Process capability is not relevant for service industries.
- E. None of the above statements is true regarding process capability.



In Six Sigma methodology, the term Critical-to-Quality (CTQ) refers to:

- A. Elements of a process that significantly affect the quality of the output
- B. The maximum number of defects allowed in a production batch
- C. A financial method used to reduce manufacturing costs
- D. The speed at which a production line operates
- E. A tool used to calculate the number of defects per million opportunities



A manufacturing process produces a product with specification limits 20 ± 1 mm. The process mean is 19.8 mm and the standard deviation is 0.2 mm. What is the process capability index C_{pk} .

- A. 0.67
- B. 1.00
- C. 1.33
- D. 1.67
- E. 2.00



The process capability index (Cpk) is used to evaluate how well a process can meet specification limits. Which of the following statements correctly describes a process with $Cpk = 1.20$?

- A. The process has a high risk of defects and is not capable
- B. The process has a moderate risk of defects and improvement is needed for better capability
- C. The process has a low risk of defects and is capable but may still be improved
- D. The process has minimal risk of defects and is highly capable
- E. The process produces zero defects



Which of the following Cpk values indicates a highly capable process with minimal risk of defects?

- A. 0.85
- B. 1.10
- C. 1.30
- D. 1.70
- E. 1.00



Six Sigma improvement methodology follows the DMAIC framework consisting of five main phases. Which of the following is NOT one of the Six Sigma DMAIC phases?

- A. Improve
- B. Define
- C. Analyze
- D. Calibrate
- E. Measure



I welcome your confidential feedback, comments,
and suggestions on the lectures I have taught.
Please feel free to share your thoughts via Moodle.

If you have any questions after the lectures, don't hesitate to reach out anytime:
Email: Amir.Parnianifard@glasgow.ac.uk

You can also reach out during office hours for an in-person chat:
Office: [Research Building 6. Room No.374, 3rd floor.](#)

**I am here to support your understanding and academic journey.
Best wishes for a successful semester!**

Dr. Amir Parnianifard

PhD, PMP®, IEEE Senior Member, RAEng Global Talent Endorsed

Assistant Professor, Glasgow College UESTC.

Email: Amir.Parnianifard@glasgow.ac.uk

Research Areas

Engineering Optimization

Computational Intelligence

Optimal Control

Uncertainty Quantification

Robotics

Wireless Sensor Networks



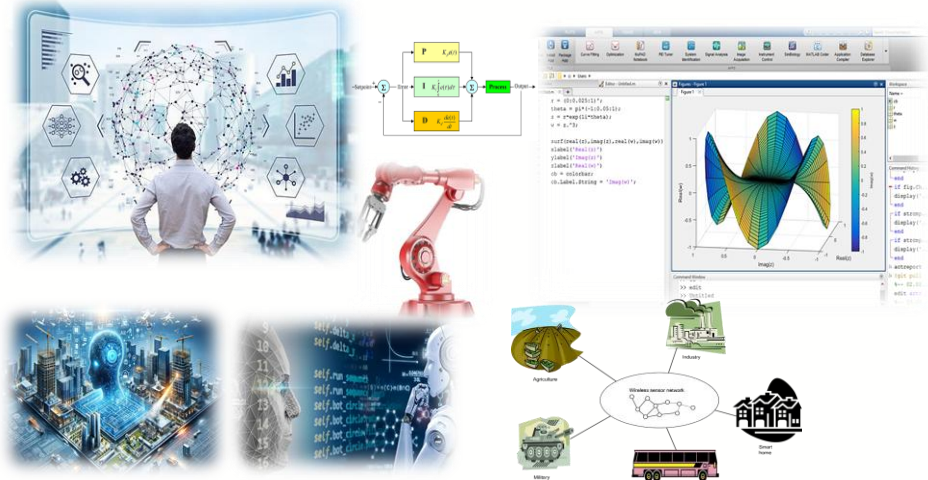
My Personal and Scholarly Page Links:

Google Scholar: <https://sites.google.com/view/amir-parnianifard/home>

ORCID-ID: <https://orcid.org/0000-0002-0760-2149>

Researcher ID (WOS): [F-7887-2017](https://orcid.org/0000-0002-0760-2149)

UofG: <https://www.gla.ac.uk/schools/engineering/staff/amirparnianifard/>



Students Advising : 23 FYP students at Glasgow College (2023-24, 2024-25, and 2025-26), and 4 postgraduate students.

Journal Publications: >50 articles, books, and conference proceedings.